Scipion-chem course script

This script follows the Scipion-chem course from a hands-on perspective, explaining step by step the process to create and manage the workflows managed in the course.

For a more theoretical explanation of the described processes, check the “Scipion-chem course documentation” file.



Agenda

	10 nov	11 nov	12 nov
9:00			
		Review (DH)	Review (DH)
10:00	Introduction to VDS (DH)	Ligand filtering (BP)	Molecular Dynamics (JA)
	Introduction to Scipion (DH)		
11:00	**Coffee break**	**Coffee break**	**Coffee break**
	Simple VDS (DH)	High-throughput screening (BP)	Atomic Modeling (BP)
12:00			
13:00	**Lunch Break**	**Lunch Break**	**Lunch Break**
14:00	Selective VDS (DH)	Flexible Docking (JA)	Protein-Protein Interactions (JM)
15:00		Lead Optimization (JA)	
16:00	**Coffee break**	**Coffee break**	**Coffee break**
	Ligand Filtering (BP)	Molecular Dynamics (JA)	One-to-one tutor meetings
17:00			
	End of day	**End of day**	**Closing remarks**

Contacts

Lead contact:
Carlos Oscar Sorzano | email: coss@cnb.csic.es

Administrative Contact:
Blanca Benitez | email: blanca@cnb.csic.es

Tutors

Daniel Del Hoyo (DH) – PhD student, Scipion-chem head developer, BCU
Blanca Pueche (BP) – PhD student, Scipion-chem developer, BCU
Joaquín Algorta (JA) – PhD student, Scipion-chem developer, BCU
Judith Maestro (JM) – PhD student, Scipion-chem developer, BCU
Laura Pérez (LP) – PhD student, Scipion-chem developer, BCU
Carlos Oscar Sorzano (COS) – Group leader, BCU; Technical Director, I2PC.

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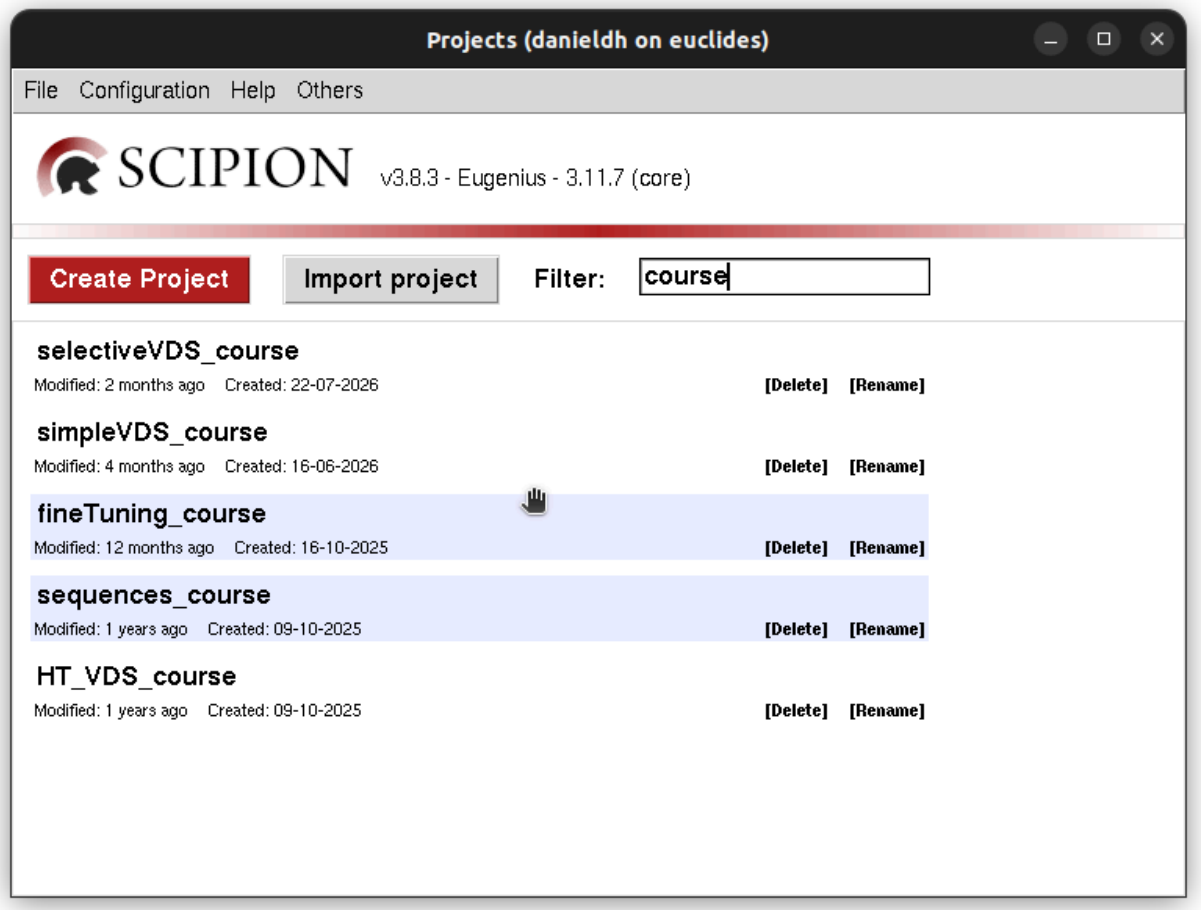
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Simple Virtual Drug Screening project

The first project aims to help you understand how to use the Scipion framework with a naive Virtual Drug Screening workflow.

1) Open the project

- **Open a new terminal ->** " Ctrl + Alt +T "
- **Invoke scipion3** to open the project manager -> " $ scipion3 "



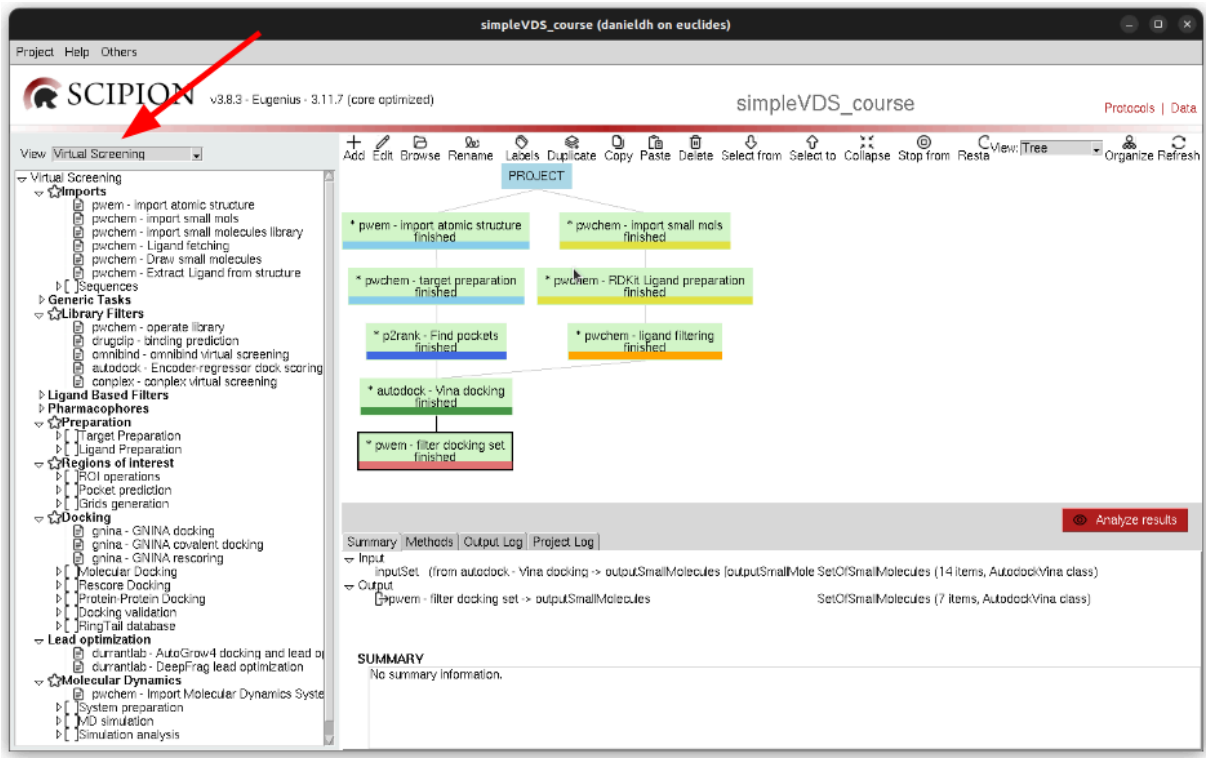
A window similar to this figure should pop up showing you the different course projects we will be using during this course. Here, you can either click on:
1) **"Create Project":** opens a new empty project window where you can create a new workflow.
2) **"Import project":** imports a project saved in a location Scipion does not find automatically and opens it.
3) **Click on the desired project** (e.g: simpleVDS_course) to open the project window.

To follow the course, **click on simpleVDS_course** and the following window should pop up.

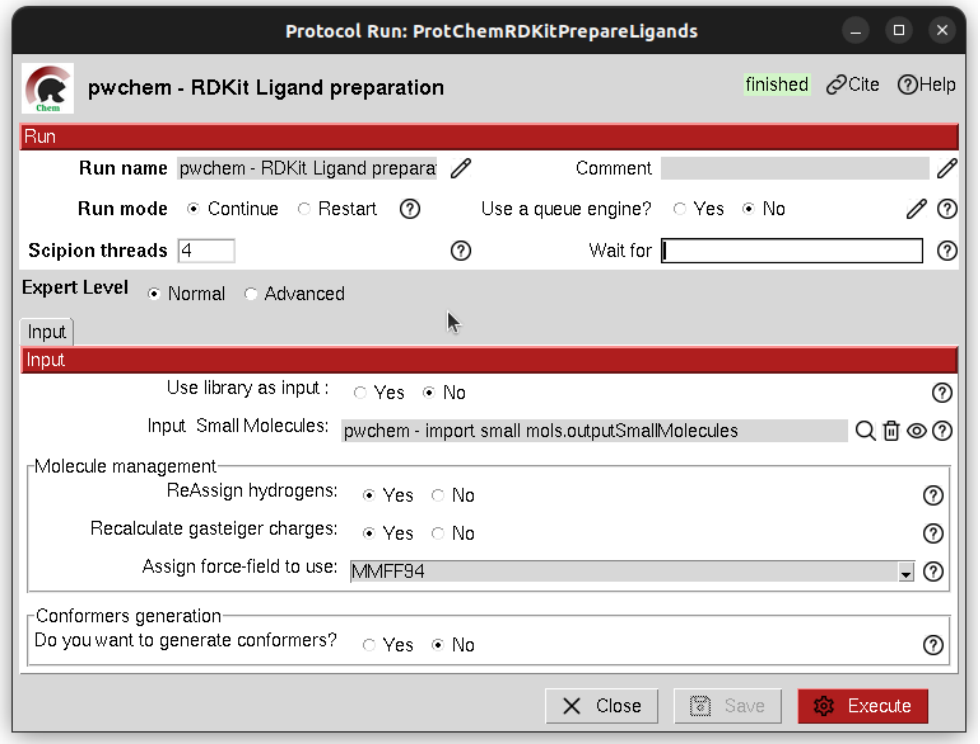
2) Inspect the project

In the project, you can visualize the workflow as a set of connected boxes, which we call protocols.

* Make sure that the **“View” section** in the top left part is set to **“Virtual Screening”** to see the related protocols. (A day may come when we want to work on “Protocols SPA”, but it is not this day).



You can edit the workflow by, for example:
1) **Creating a new protocol** from the list in the left panel or look for it by name using “Ctrl+F”.
2) **Opening an existing protocol** by double clicking on its box. It will open the protocol form (try the “RDKit ligand preparation).



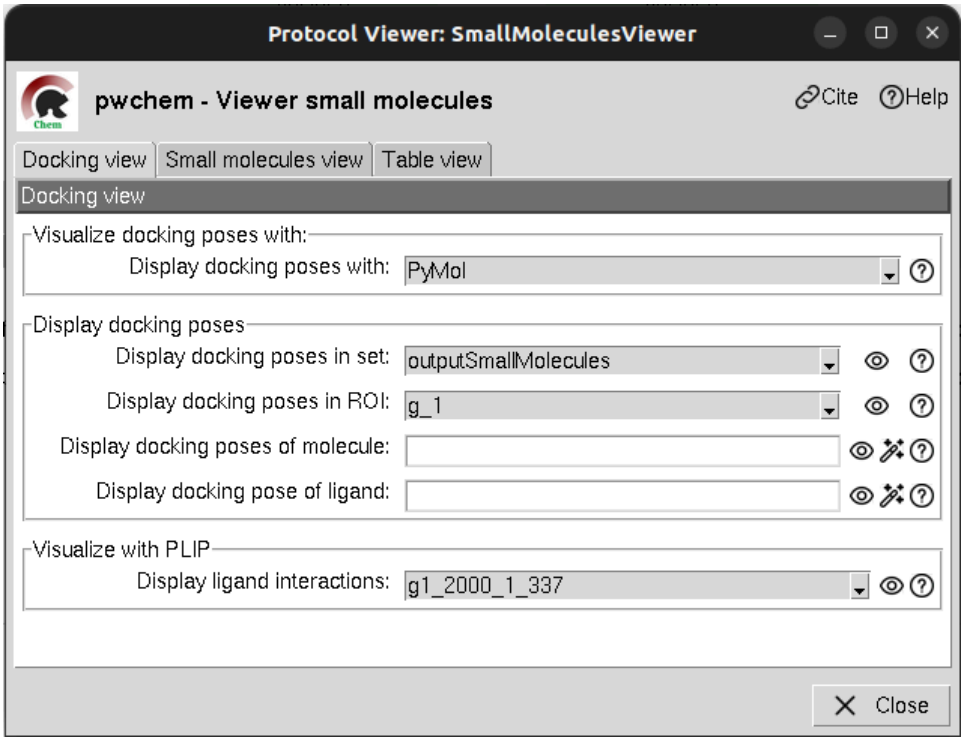
The **form window** shows the **parameters and inputs** the protocol has been run with.

Options:

- **Restart** the protocol by setting **"Run mode" to "Restart"** and clicking on **"Execute".** However, notice that this will delete the previous execution of this protocol and its data, together with the ones of the downstream workflow, so use with care.
- **Duplicate the protocol,** which will let the existing protocol as is, and create a **new box** as a copy of it, where you can change the parameters. To do so, select the box and press **"Ctrl+D"** or double click on it and select "**Duplicate**". Once you click on "**Execute**", the new box will appear and the protocol will run within the new box.

3) Explore the output

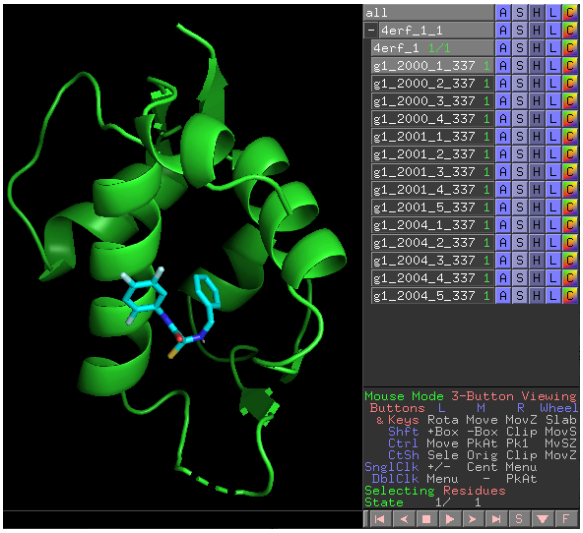
Using the **"Analyze results"** button on the project window (bottom-right), you can explore the **output of the selected protocol.** Depending on the kind of output, a visualization will directly pop up or, more likely, a viewer form will be opened showing you different viewer options. For example, try analyzing the "Vina docking" output:

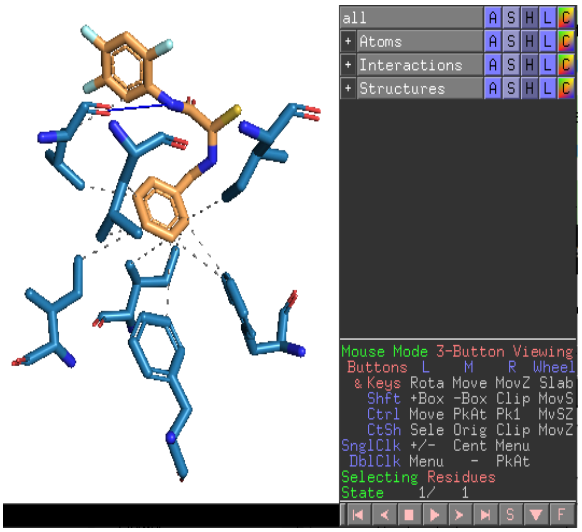


It includes 3 sections with different **visualization modes** of the output molecules:

1) **Docking view**: plots the docked molecules onto the receptor.

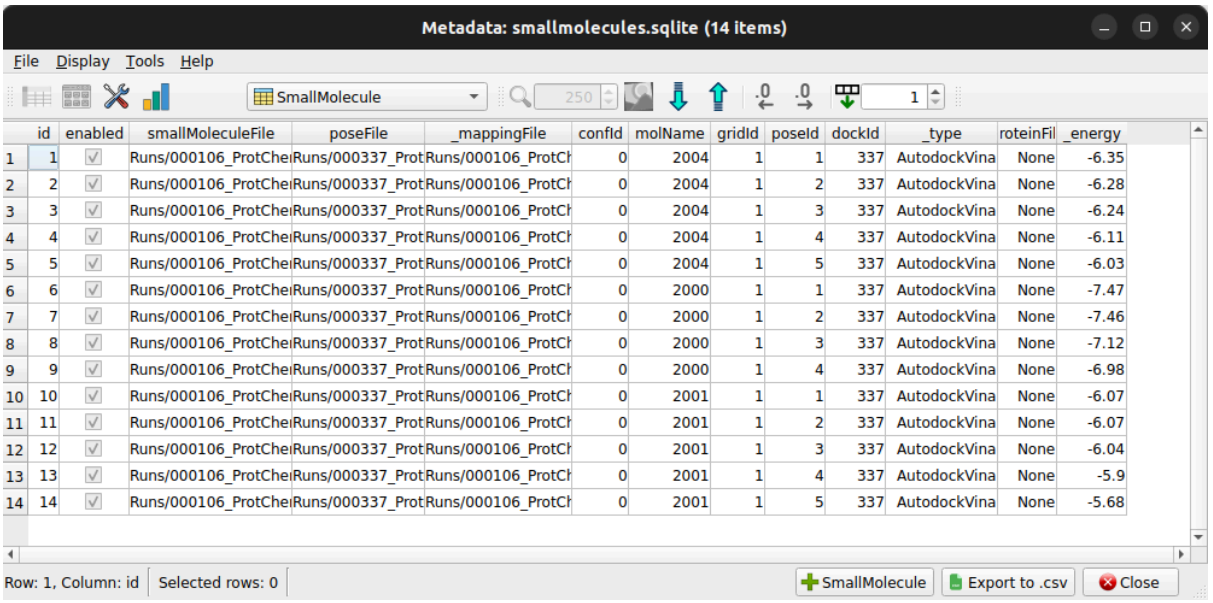
- You can visualize them on PyMol, ChimeraX or ViewDockX (a special ChimeraX mode).
- You can choose whether to display the molecules by **set** (all molecules in this case), by **ROI** (Region Of Interest, protein pockets in this case), by **molecule** (all poses of a certain molecule) or a **single pose**. Select the desired option and click on the corresponding **eye icon** to open the viewer. You can use the **wand icon** to help you select the desired options.
- A final option uses **PLIP,** a special PyMol option to show a single ligand pose plotting all its found **interactions with the receptor**.





2) **Small molecules view:** plots the molecules prior to the docking, without the receptor. It contains similar options to those in the “Docking view”

3) **Table view**: displays a table where **each element of the output set is represented as a row** and where the **columns contain different features** of those elements. In this case, each row is a molecule pose and you can check features like the docking energy, the docking software used or the molecule name.



	id	enabled	smallMoleculeFile	poseFile	_mappingFile	confId	molName	gridId	poseId	dockId	_type	roteinFil	_energy
1	1	☑	Runs/000106_ProtChe	Runs/000337_Prot	Runs/000106_ProtCh	0	2004	1	1	337	AutodockVina	None	-6.35
2	2	☑	Runs/000106_ProtChe	Runs/000337_Prot	Runs/000106_ProtCh	0	2004	1	2	337	AutodockVina	None	-6.28
3	3	☑	Runs/000106_ProtChe	Runs/000337_Prot	Runs/000106_ProtCh	0	2004	1	3	337	AutodockVina	None	-6.24
4	4	☑	Runs/000106_ProtChe	Runs/000337_Prot	Runs/000106_ProtCh	0	2004	1	4	337	AutodockVina	None	-6.11
5	5	☑	Runs/000106_ProtChe	Runs/000337_Prot	Runs/000106_ProtCh	0	2004	1	5	337	AutodockVina	None	-6.03
6	6	☑	Runs/000106_ProtChe	Runs/000337_Prot	Runs/000106_ProtCh	0	2000	1	1	337	AutodockVina	None	-7.47
7	7	☑	Runs/000106_ProtChe	Runs/000337_Prot	Runs/000106_ProtCh	0	2000	1	2	337	AutodockVina	None	-7.46
8	8	☑	Runs/000106_ProtChe	Runs/000337_Prot	Runs/000106_ProtCh	0	2000	1	3	337	AutodockVina	None	-7.12
9	9	☑	Runs/000106_ProtChe	Runs/000337_Prot	Runs/000106_ProtCh	0	2000	1	4	337	AutodockVina	None	-6.98
10	10	☑	Runs/000106_ProtChe	Runs/000337_Prot	Runs/000106_ProtCh	0	2001	1	1	337	AutodockVina	None	-6.07
11	11	☑	Runs/000106_ProtChe	Runs/000337_Prot	Runs/000106_ProtCh	0	2001	1	2	337	AutodockVina	None	-6.07
12	12	☑	Runs/000106_ProtChe	Runs/000337_Prot	Runs/000106_ProtCh	0	2001	1	3	337	AutodockVina	None	-6.04
13	13	☑	Runs/000106_ProtChe	Runs/000337_Prot	Runs/000106_ProtCh	0	2001	1	4	337	AutodockVina	None	-5.9
14	14	☑	Runs/000106_ProtChe	Runs/000337_Prot	Runs/000106_ProtCh	0	2001	1	5	337	AutodockVina	None	-5.68

4) Create your own project

Go back to the **project manager window**, **create a new project** and try to r**ecreate the workflow** we have just analyzed.
Remember to set the **“View” as “Virtual Screening”** to visualize the proper protocols in the left panel or use **Ctrl+F** to find the desired protocols.

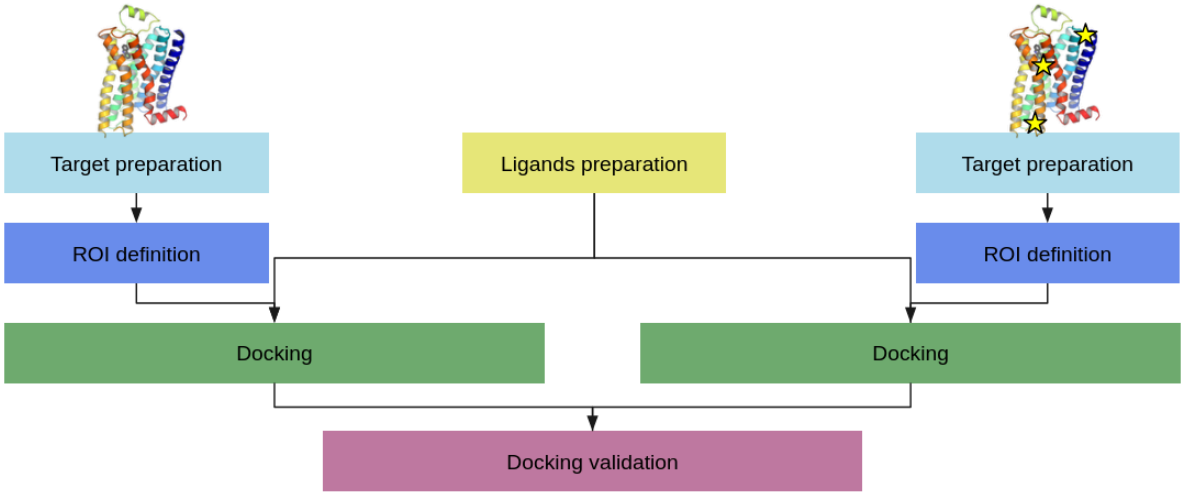
The first protocols (Import atomic structure / small mols) create the **Scipion objects** (AtomStruct / SetOfSmallMolecules) that downstream protocols will use as input. Use the **magnifying glass icon** to select the input of a protocol among the **Scipion objects** present in the project.

Selective Virtual Drug Screening

This second project contains a workflow that runs 2 parallel screenings for 2 homologous PI3K (Phosphatidyl Inositol 3-kinases), α (alfa) and δ (delta) over the same set of molecules. These molecules are known to be bound differentially to these 2 protein isoforms, so the objective of this project is to test whether we are able to recover those differences using in-silico predictions.

1) Inspect the project

From the project manager window ("$ scipion3" on terminal) open the **"selectiveVDS_course"** project.
The project is basically divided into 3 branches that merge on the downstream steps of the workflow: a first branch for PI3Kα, another for PI3Kδ and a third one for the ligands, as illustrated in the following image.



Notice the color codes that are also present in the workflow protocols to facilitate the visualization.

2) Explore the workflow steps

2.0) Imports and preparations

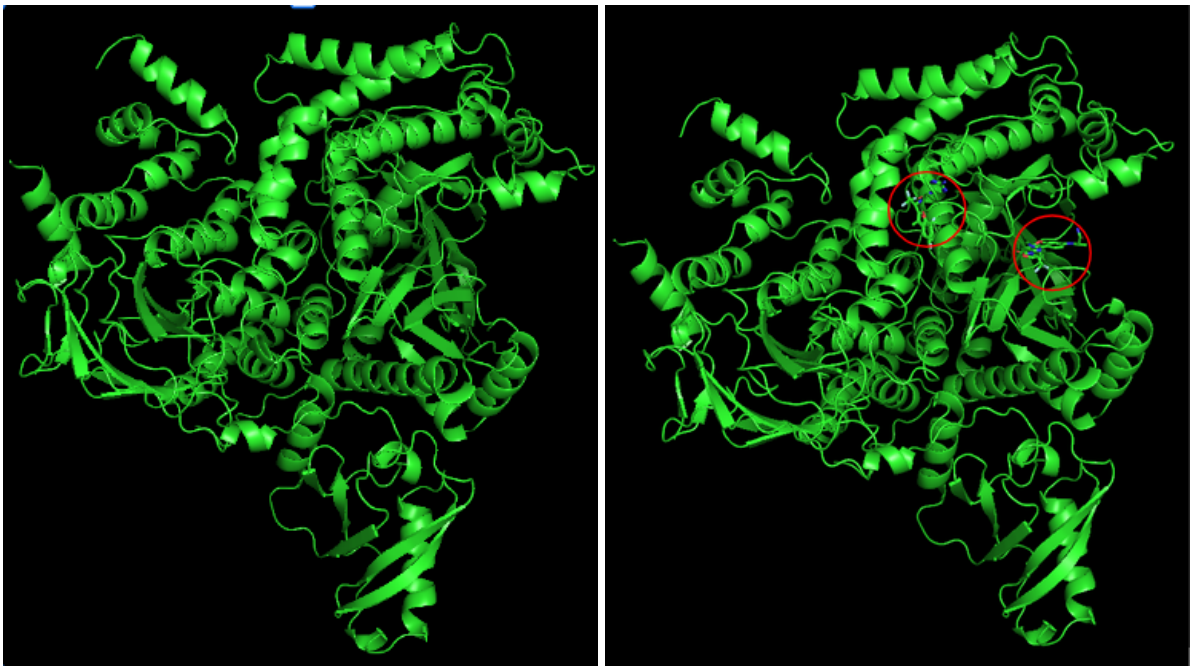
The first step for any workflow is usually importing the structures we are going to use and prepare them according to our needs.

a) **PI3K structures**:

- Structures for PI3Kα (8TDU) and PI3Kδ (8BCY) are **downloaded from PDB**. Local structure files could also be imported.
- **Receptor:** the receptor is built by removing the existing ligands and keeping only chain A.

- **Complex:** here, not removing the ligands allows us to observe the interactions with the receptor and compare them to the ones we will predict.

The results of the receptor and the complex generation protocols are pretty similar (see next figure). The only difference is that the complex still has the ligands present in the original structure.



b) <u>PI3K ligands:</u>

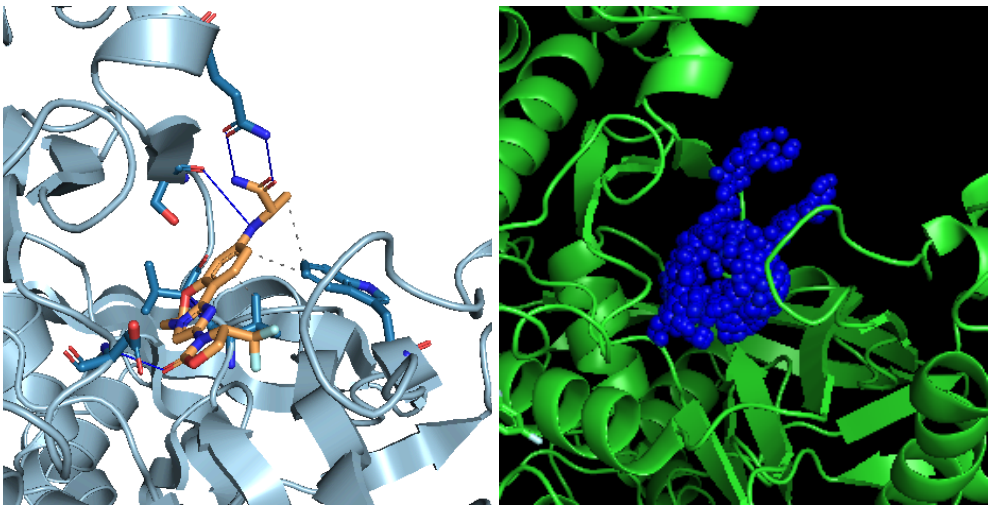
- Downloaded from **PubChem** codes or from local files.
- Preparation generates their **3D structure conformers** from 2D or SMILEs.

2.1) Druggable pocket analysis

The next section is used to study and describe the binding pocket where the original ligands were attached to.

a) The **complex** can be used to:

- **Extract the ligand:** visualize the interactions as if it was docked
- **Define the pocket:** as the residues next to the ligand

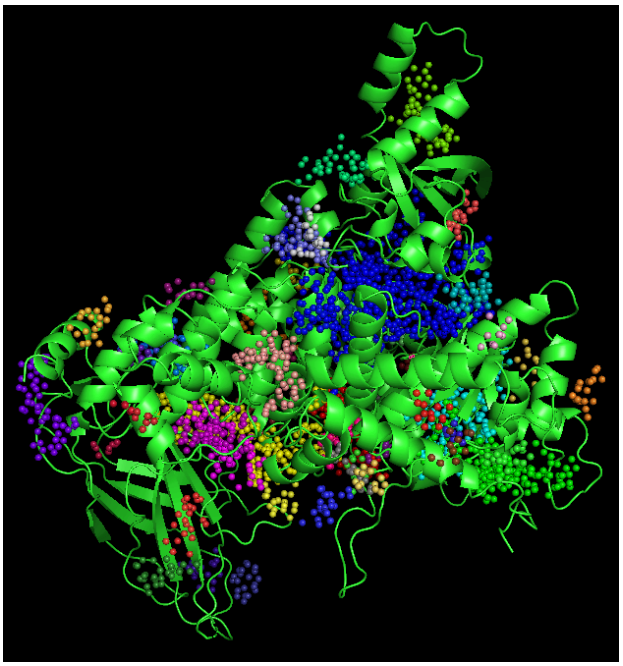


The figure shows the PLIP visualization of the ligand interactions (left) and the inner surface of the defined pocket covered by beads.

b) **Analyze the receptor:** you can perform analysis on the receptor such as calculating the **SASA**.

c) Pocket prediction: we can predict the pockets over the receptor structure. If the pocket was unknown, we may use these predictions for docking. In this case, we will just check that the correct pocket is included.

- **Predictions** using **P2Rank, FPocket** and **AutoSite**



This kind of predictions tend to have a lot of pockets, most of them are usually of poor quality. To filter those pockets, we can apply different strategies. The most direct uses the **scores** associated with the predictions to **filter those of lower quality.** Another option provided by Scipion involves the **consensus protocol**, that outputs those pockets that are present in several inputs (in this case, the inputs would be the predictions).

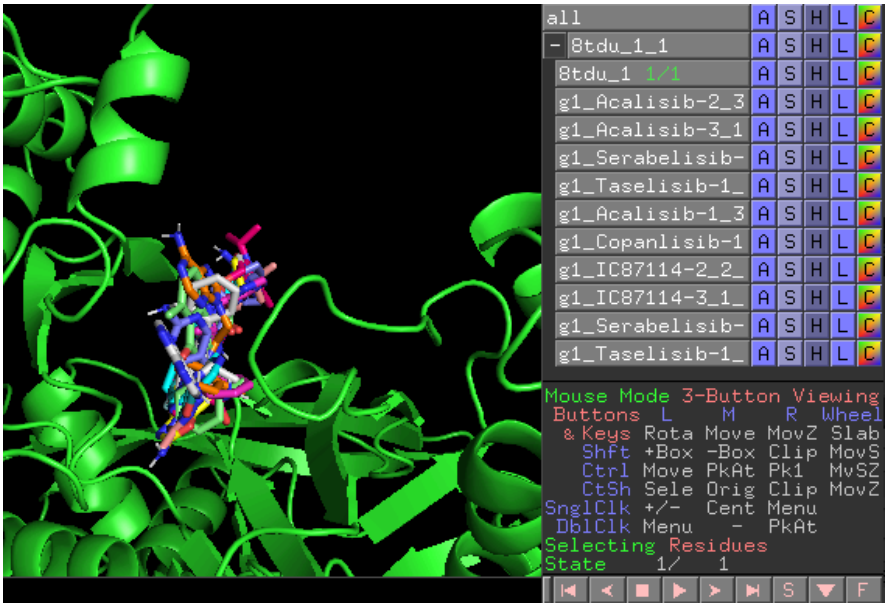
- **Consensus pockets** over predictions: define those that are shared

- **Consensus with real pocket:** check if they are similar

2.2) Docking analysis

Once we define the pocket(s) where we want to check whether our molecules can be attached, we use docking protocols to prove different likely positions and calculate an associated score.

a) **Docking:** we use AutoDock-GPU, VIna and LeDock to have a variable set of docking outputs. Similar to the case of the pockets, those positions can be filtered in several ways. The most direct would be applying the calculated scores and filtering the worse scoring options. In Scipion, we add another consensus protocol that looks for repeated positions from different inputs.



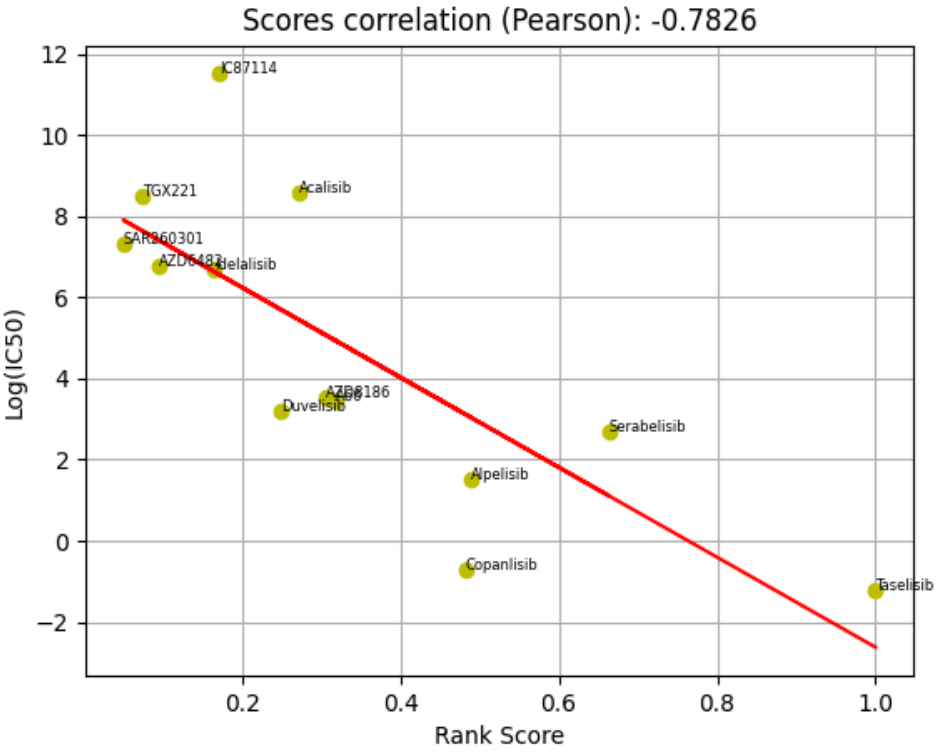
b) **Consensus docking:** it builds **clusters of similar positions** and outputs a representative of those clusters sufficiently populated. The **representative** is usually chosen as that **pose with a better score**, but scores of different software are not comparable, so a **rescoring** is needed first.

2.3) Docking validation

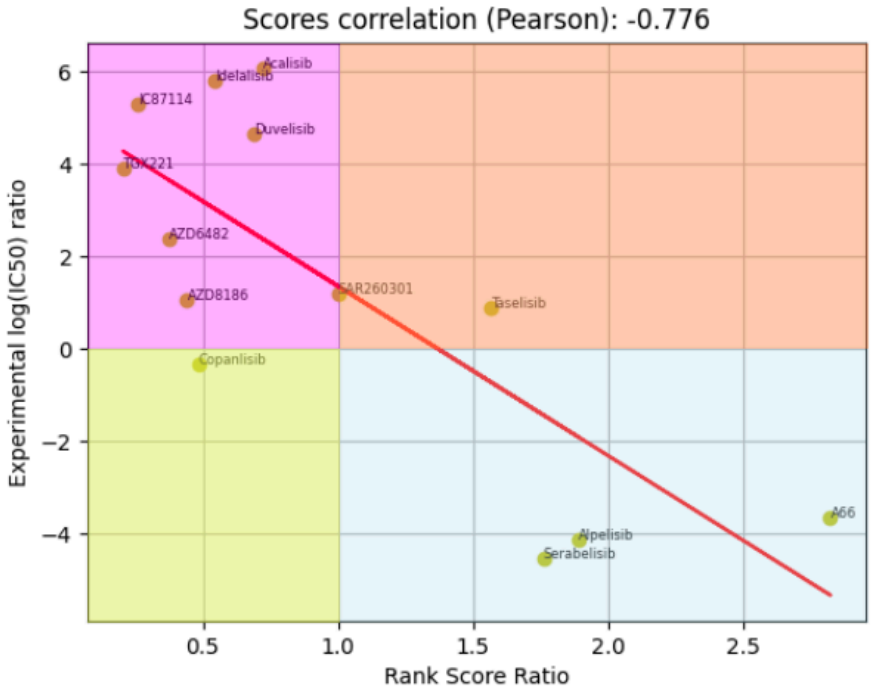
In this section, the docking results for each docking software plus the consensus are evaluated against the actual affinity values for each molecule.

a) **Ranking score:** it's an algorithm to aggregate the docking scores for every docking pose and conformers that have been generated during the workflow onto a single score for each molecule. Another strategy would imply taking the best docking score for each molecule.

b) **Score correlation:** used as validation of each **VDS** (and each docking method) workflow. We checked the **Pearson correlation** of the actual affinity values with the scores obtained for each molecule. Experimental affinity values (IC50) are transformed via log10 to be more accessible.



c) **Score ratio correlation:** used to validate the **selective VDS** workflow as a whole. In this case, we calculate the **Pearson correlation of the ratios** of the actual and the predicted scores for each molecule. This way, we checked whether we are able to recover the selective information.



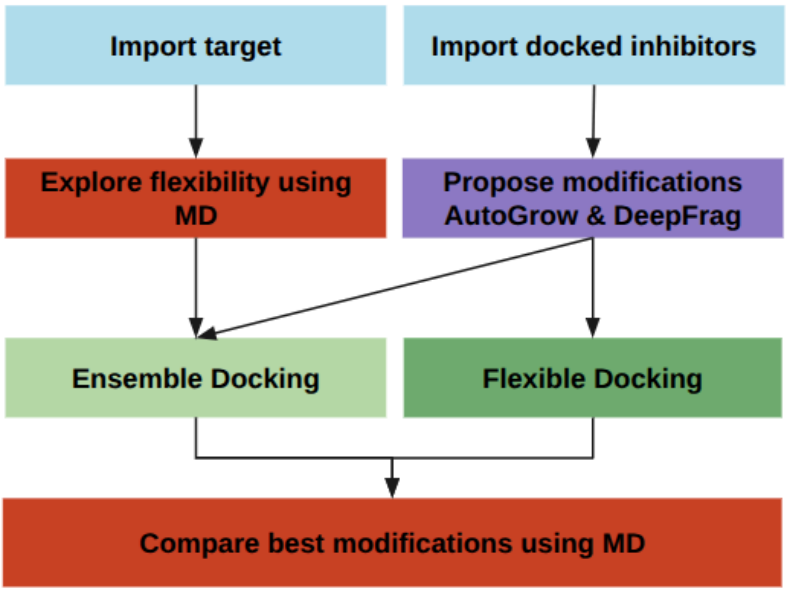
Fine-Tuning Virtual Drug Screening

The first two projects screened a fixed set of molecules against a fixed target structure. This third project relaxes both: the molecules are not fixed, and neither is the receptor. We start from compounds already known to bind PI3Kα, generate new ones from them, and then spend real computing time deciding whether the new ones are actually better.

The objective of this project is to generate molecules from the set of inhibitors used in the Selective Virtual Drug Screening project that increased the affinity to PI3Kα. There will be used **Lead optimization**, **Flexible docking** and **Molecular dynamics** protocols.

1) Inspect the project

From the project manager window ("$ scipion3" on terminal) open the **"FineTunePI3Ka_course"** project.



The project is divided into 2 branches:

- Right branch — Lead optimization: generate new molecules from the known inhibitors with AutoGrow4 and DeepFrag, then dock them with a flexible receptor.
- Left branch — Ensemble docking from a molecular dynamics trajectory: simulate the apo target, extract representative conformations, and dock against all of them instead of a single structure.

Both branches produce docked candidates, and the final section compares the best pose from each docking approach against the original inhibitor using molecular dynamics.

Note that the colors match the Scipion labels colors.

2) Explore the workflow steps

2.0) Different imports

Import PI3Kα, which will be used again as target. We perform 2 preparations of it: one removing the ligands, used to explore the target flexibility by molecular dynamics, and another keeping the ligands, used to define the pocket.
For the lead optimization we import two sets of molecules:

1. The exported consensus docking results for PI3Kα from the selective VDS project ("selectiveVDS_course"). These are the **known inhibitors**, already docked, and they are the starting points for both generation tools.
2. A **fragment library** from the ZINC database (https://zinc.docking.org/) containing fragments under 100 Da, already downloaded.

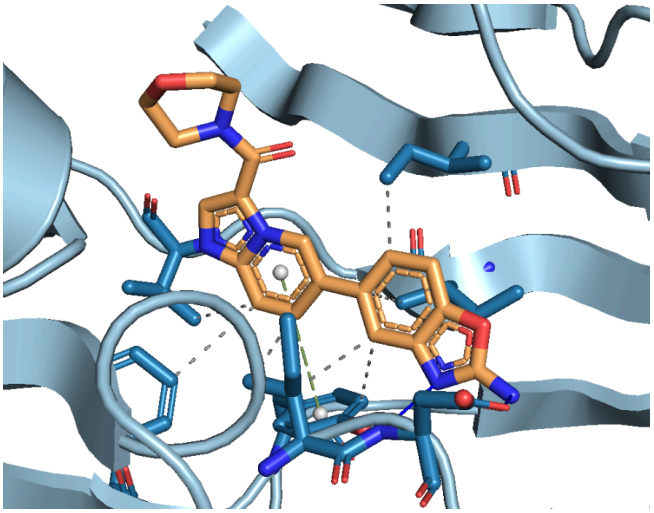
2.1) Lead Optimization

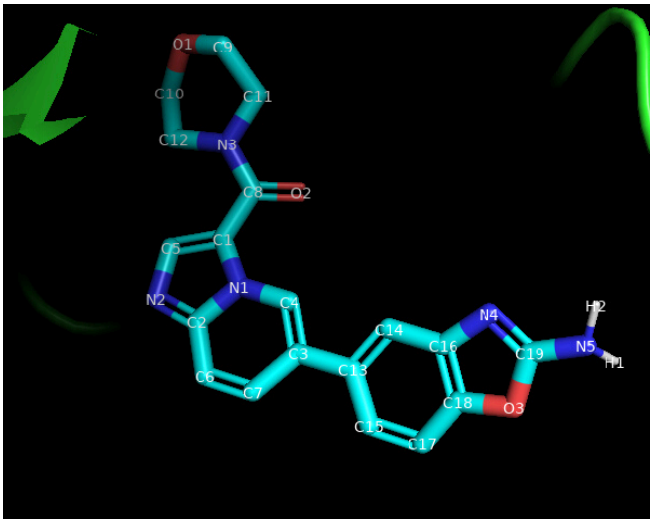
For Lead Optimization we use AutoGrow4 and DeepFrag2.

Autogrow needs a pocket and a molecule library to make several rounds of optimization. There are 2 different AutoGrow runs:

1. Using only the ZINC fragment library.
2. Using a joint set of the PI3Kα inhibitors and the fragment library.

We do this because AutoGrow needs a relatively big set for doing its operation, a minimum of hits every generation is needed.





DeepFrag is more direct: it needs a pose, and the user defines the atoms from which it will grow or substitute fragments. To choose good atoms, use the two viewers together. The PLIP viewer (left figure) shows which atoms are already involved in interactions, so you can avoid modifying those. The DeepFrag viewer (right figure) displays the atom numbering, so you can identify the atom names to enter in the form.

We chose to perform 2 different runs, both using the Serabelisib-1_2 pose as input:

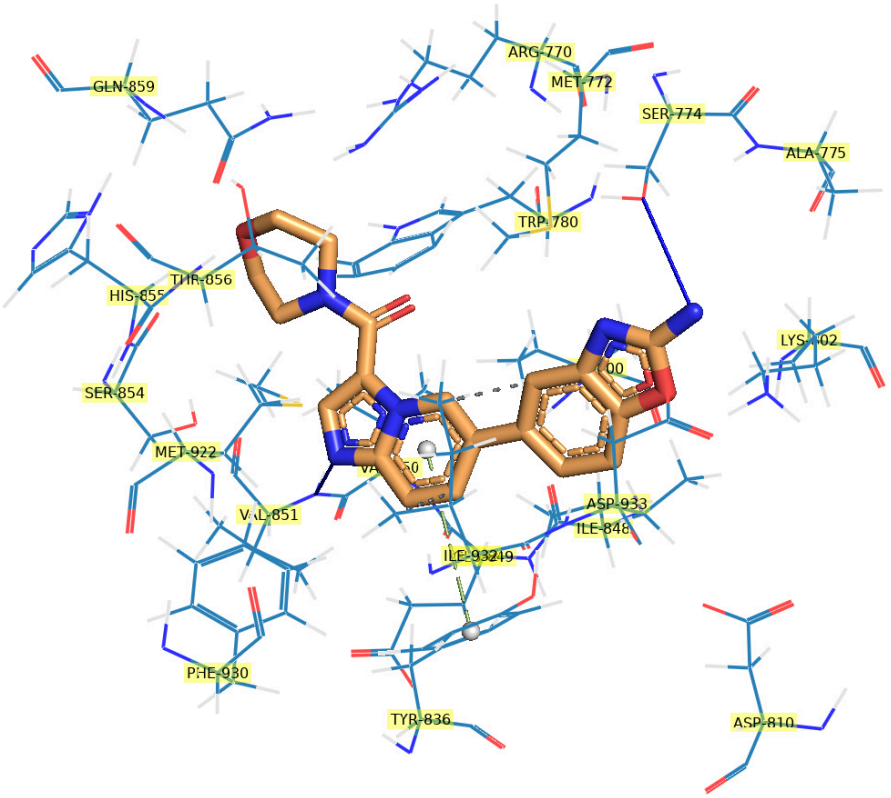
1. One that grows fragments from atoms C11 and C10.
2. One that substitutes the fragments attached at bonds C1–C8 and C19–N5.

AutoGrow generates docked poses but DeepFrag does not. After filtering the AutoGrow poses by energy (keeping those below −6.5 kcal/mol), we join the 4 resulting setOfSmallMolecules, 2 from AutoGrow and 2 from DeepFrag, into a single set, which is then used for docking.

2.2) Flexible Docking

After preparing the ligands we are going to perform flexible docking using Autodock. This protocol has the option to select specific residues from the target that will be treated as flexible (flexible bonds). This increases the search space a lot. For that reason it is recommended to **select only a few residues**, and to **increase the search exhaustiveness** parameters to compensate: a larger space searched with the same effort gives worse results than a rigid docking, which is an easy way to conclude wrongly that flexibility does not help.

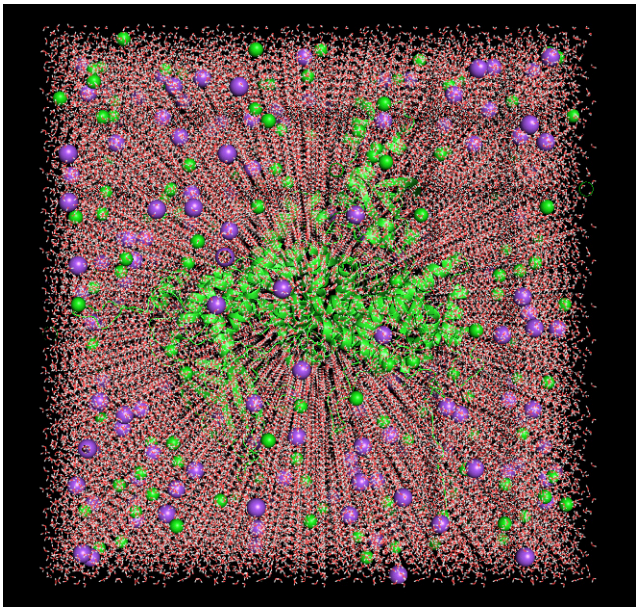
Which residues to make flexible is a decision for the user, based on the pocket. It is recommended to choose residues that may be blocking the site, or whose flexibility could generate additional interactions. Use PyMOL to inspect the pocket and one example pose, as in the next figure. In our case, we selected MET922 and GLN859 because it covers the part of Serabelisib where we are growing the molecule using DeepFrag.



Results of Flexible docking are very similar to rigid docking results. However, when opening Pymol we can see one target per generated pose that accounts for the moving residues.

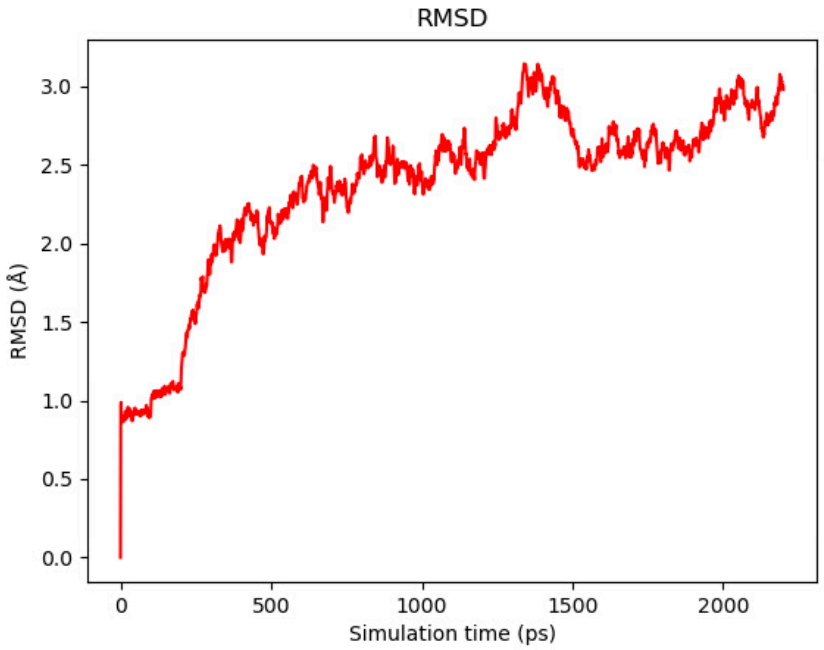
2.3) Molecular Dynamics for PI3Kα flexibility

From the clean PI3Kα we perform a simulation to see whether the pocket moves. Before simulating, the system must be prepared with the "**system preparation**" protocol. We add a salt concentration of 0.15 M and keep the other parameters at their defaults, resulting in a solvated protein.



For the **simulation** we are performing 5 steps protocol: Minimization, NVT of 100ps, NPT of 100ps, NPT of 1000ps and NPT 1000ps. The first 3 have the Backbone fixed and the 2 next are unbiased. In order to see when the simulation equilibrates, we can plot the RMSD of All stages.

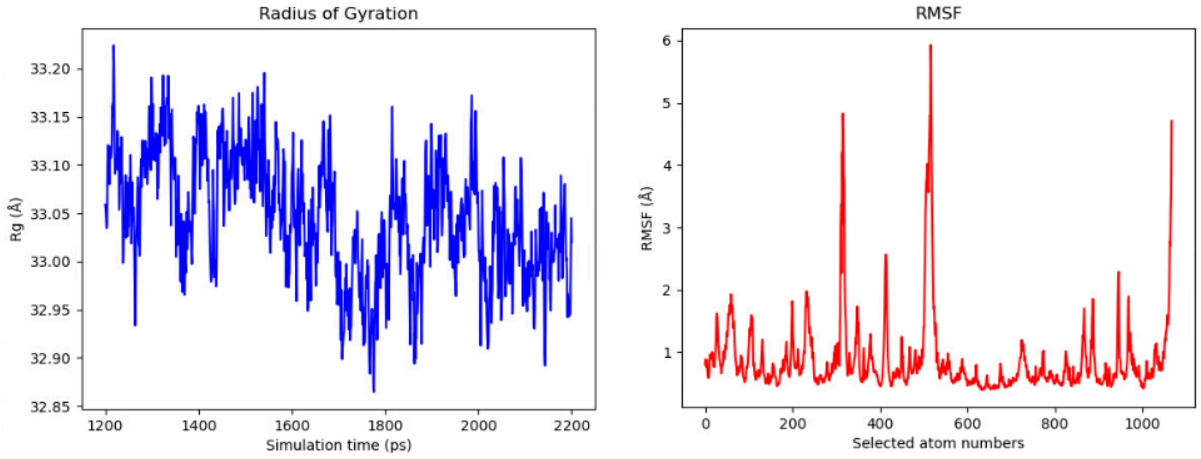
Checking equilibration. Plot the RMSD of all stages together. In the first NVT and NPT stages (0–200 ps) the protein moves a lot. In the first unbiased NPT (200–1200 ps) the RMSD keeps rising. The last NPT looks equilibrated: it still moves up and down, but around a stable value.

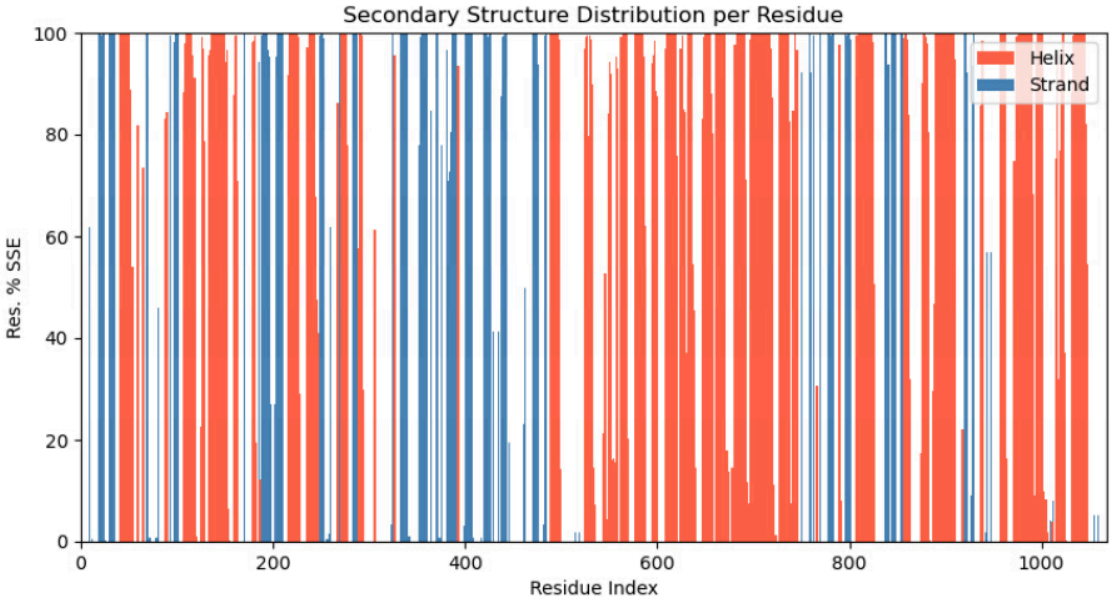


Use the "**system modification**" protocol to cut the trajectory and keep only the equilibrated part (1200–2200 ps). Everything below is computed on that window.

Analysing the flexibility: In the viewer there are a lot of options to study the flexibility of the protein. For example:

1. Radius of gyration confirms the fold stayed intact. Rg measures global compactness and over this 1 ns window the total variation is about 0.4 Å, between roughly 32.85 and 33.25 Å, so the protein remains compact and globular throughout.
2. RMSF measures which atoms fluctuate most, and this is where the pocket is assessed. We are interested in the flexibility of the binding-pocket residues (770, 772–775, 778, 798, 800, 802, 805, 807, 810, 814, 834, 836–838, 848–856, 859, 919, 922, 930–934, 936). If we plot with the CA option residues around 800 are quite stable, but those around 850 show a peak of about 2 Å.
3. Secondary structure plot per residue shows that the pocket combines strands and helices with some short loops, which is consistent with the RMSF being concentrated in the 850 region.



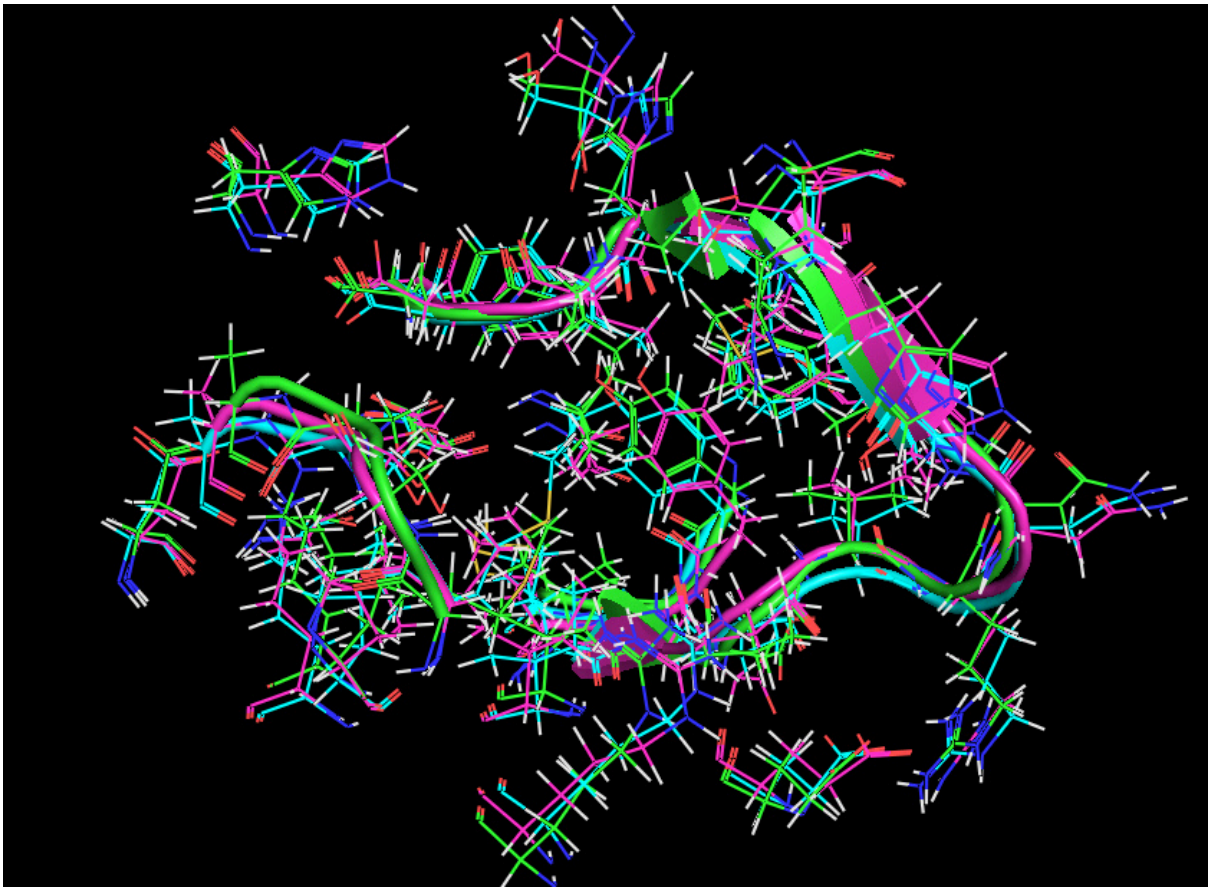


So the fold is stable while part of the pocket is not, and the side chains are rearranging.

2.4) Ensemble Docking

The idea of ensemble docking is not to use a single target structure but several. Here we build clusters from the MD simulation and run an independent docking against each representative, which is how the target flexibility measured in 2.3 is carried into the docking.

"**Trajectory clustering**" protocol extracts representative structures of a trajectory based on RMSD between each frame. We predict pockets using p2rank from each representative structure and select the pocket corresponding to the studied pocket (this selection is manual by looking at the contact residue from the table viewer). By viewing the cluster structures and showing the pocket residues we can see how the flexibility of the sidechains are represented.



To those 3 pockets we perform docking using Autodock-GPU from the generated lead optimization set. Then, for the 3 autodock outputs, we take the best pose for each molecule using "**operate set**" with Best operation param, molName reference column and _energy as filter column. After that we will use the "**rank docking score**" protocol, which creates a new score based on the ranking of each molecule on each independent docking run to select our best candidates.

2.5) Binding Pose Stability using MD

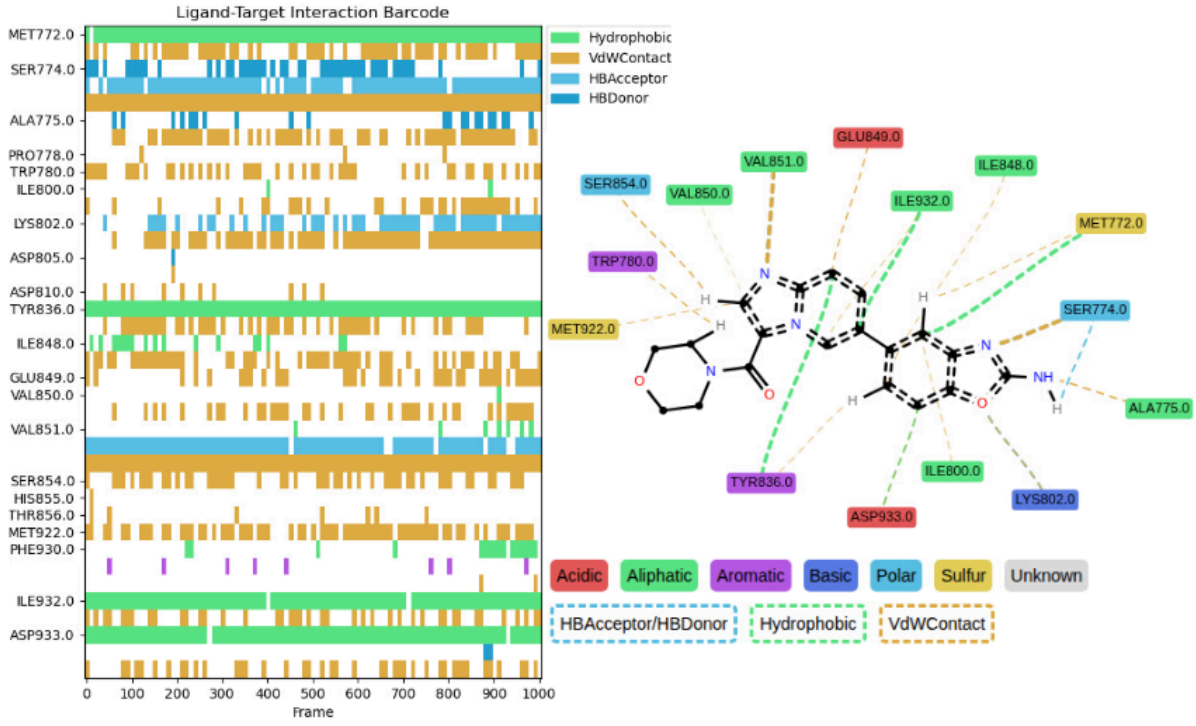
We are going to compare the binding pose stability of the input Serabelisib-1_2, the best scoring pose from the ensemble docking and the best scoring pose from the flexible docking.

As an example I will use the Serabelisib-1_2 pose. The system preparation, simulation, equilibration inspection using RMSD and the modification to take the equilibrated part of the trajectory is the same as the molecular dynamics we performed in the 2.3 section but now we take the molecule of interest from the setOfSmallMolecules as input.

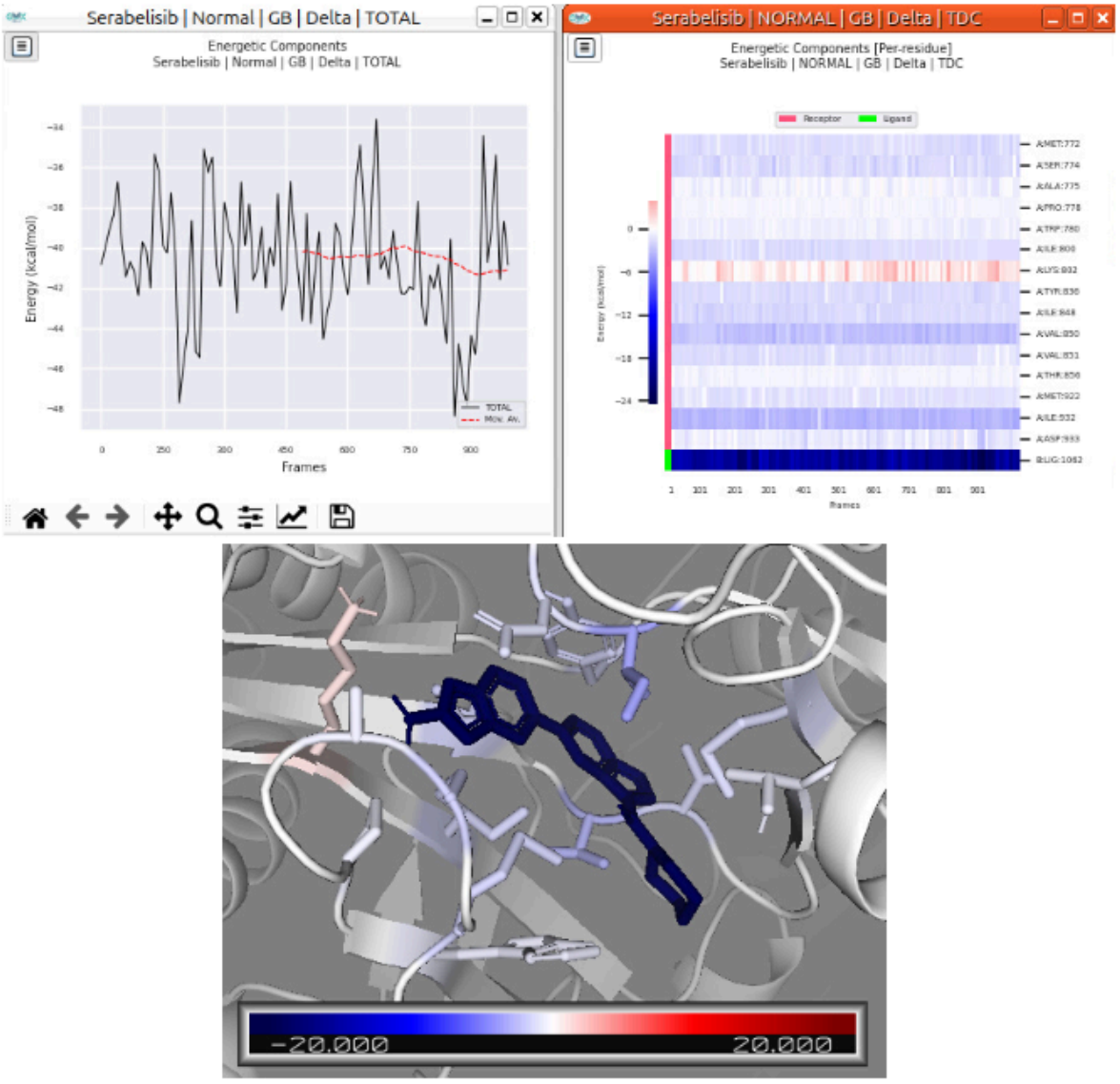
Inspecting the trajectory: The viewer gives several measures:

- Ligand RMSD - how much the ligand moves over the trajectory. A value that rises and plateaus means the ligand settled into a slightly different pose; one that keeps climbing means it is leaving the pocket.
- Protein-ligand minimum distance.
- Protein-ligand hydrogen bonds, counted over time.

Apart from that we will use “**ProLIF protocol**” (Protein-Ligand Interaction Fingerprint) to extract the specific interactions that are happening throughout the trajectory of the simulation. This way we can have an idea of how tight those unite:



To finish our binding stability analysis we can perform a binding energy estimation using MM/PBSA or MM/GBSA using the protocol (“**MM/PBSA free energy calculation**”). We choose GBSA without entropy correction because it is faster (but less precise). Apart from a final energy value shown in summary, using the viewer an interactive analysis can be open with a lot of information about how the binding energy evolves over time and which residues of the protein are the ones contributing or desestabilizing the binding. Treat these values as a way of ranking closely related molecules against the same receptor, not as absolute affinities.

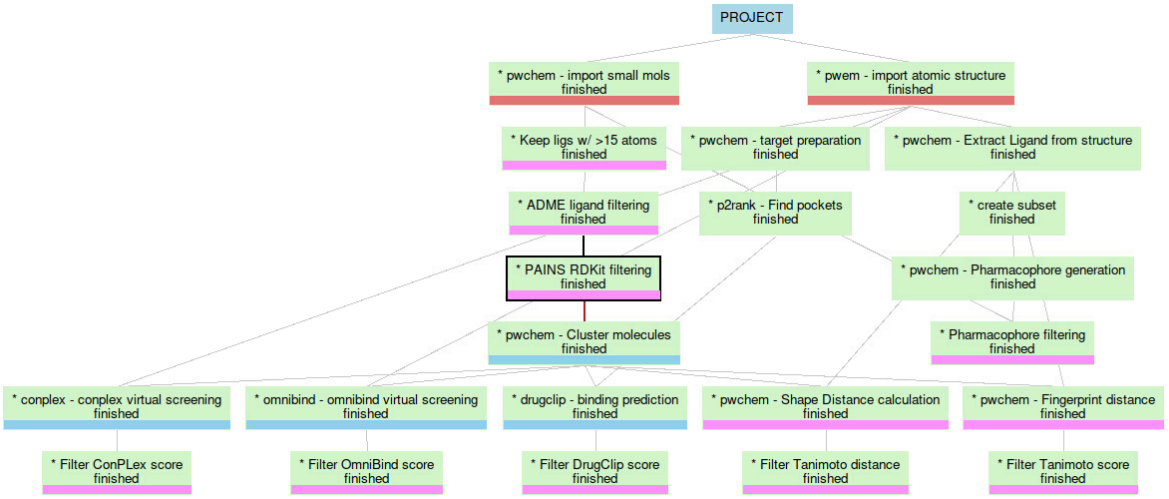


The idea is to reproduce the same Molecular Dynamics analysis to the top ranking pose of the Ensemble docking and the top ranked pose of the Flexible docking. This will show whether our lead optimization project produced molecules with higher affinity and more stable poses than Serabelisib.

Inside the project the protocols corresponding to the MD analysis of poses are named by **Ori** (Original Serabelisib-1_2 pose), **ED** (Ensemble Docking outputs from the lead optimization created molecules) and **FD** (Flexible Docking outputs from the lead optimization created molecules).

Ligand filtering and High-throughout VS

Ligand filtering aims to reduce a chemical library to a more manageable set of compounds before running more computationally expensive screening methods. In this practical, we will explore the filtering protocols available in Scipion-chem and see how different criteria can be used to select molecules according to their chemical properties and structural features.



1) Inspect the project

Open the project containing the ligand filtering workflow, called “course_workflow_filteringHTVS”.

Inspect the workflow and identify the input ligand set, the filtering protocols, and the resulting set of molecules. Pay attention to how the protocols are connected and how each step affects the number of compounds retained.

2) Explore the filtering criteria

Ligand libraries can be filtered using different types of criteria, depending on the objective of the screening campaign.

- **General filtering criteria:** removes/keeps molecules that do not/do satisfy basic structural requirements, such as restrictions on atom types, molecular size, or ring-related properties, depending on the available protocol.
- **Physicochemical property filtering:** selects compounds according to properties such as molecular weight, lipophilicity, hydrogen-bond donors and acceptors, or other drug-likeness descriptors.
- **ADMET filtering:** uses predictions related to absorption, distribution, metabolism, excretion, and toxicity to help prioritize compounds with potentially suitable properties.
- **Structural-alert filtering:** identifies compounds containing patterns associated with potential assay interference or other undesirable chemical features, such as those targeted by PAINS filters.

- **Similarity-based filtering:** selects compounds according to their similarity to reference molecules, using representations such as molecular fingerprints, shape, or pharmacophore features, when supported by the selected protocol.

Not every workflow needs all these filters. The appropriate criteria depend on the compounds, the target, and the purpose of the screening campaign.

3) Analyze the filtering output

After executing the filtering protocols, compare the input and output ligand sets.

Check:

- The **number of molecules retained** at each step.
- Which properties or structural features determine whether a molecule passes the filter.
- Whether the filters are complementary or remove overlapping groups of compounds.
- Whether the **resulting library remains sufficiently diverse** for the intended screening task.

Filtering can improve computational efficiency, but applying overly restrictive criteria may also remove potentially interesting candidates. The objective is to obtain a suitable library, not simply the smallest possible one.

In the first filtering steps (left branch of the workflow) we applied 3 main filters: keep molecules with more than 15 atoms, those that pass the LIpinski Rule of 5, and the ones that pass the PAINS filtering. From a group of 4927 molecules we ended up with 4304.

4) Explore ligand clustering

Clustering groups similar molecules to reduce redundancy and organize a chemical library. In this workflow, clustering is applied to the filtered ligand set before HTVS.

Inspect the clustering protocol and its output. Compare the number of molecules before and after clustering, and consider how grouping similar compounds can make a large library easier to explore.

In this workflow, clustering reduced the library from 4,304 molecules to 1,333 representative compounds.

5) Explore screening methods and analyze output

Scipion-chem provides access to different approaches for prioritizing compounds. The required inputs and interpretation of the results depend on the selected method.

- **Ligand clustering:** groups similar compounds to help identify redundancy and organize candidate molecules. Clustering can be used before screening to reduce redundancy or after screening to organize the predicted hits.

- **ConPLex:** uses a protein sequence and a ligand representation to predict protein–ligand compatibility through a contrastive-learning approach.
- **OmniBind:** applies a machine-learning-based approach to protein–ligand interaction prediction. Inspect the specific protocol inputs to understand which protein and ligand representations are required.
- **DrugCLIP:** uses contrastive learning to prioritize ligands based on their compatibility with a protein binding pocket.

These methods provide complementary ways of prioritizing compounds. They do not necessarily need to be run sequentially, and their scores should not automatically be interpreted as experimentally measured binding affinities.

Explore the results produced by the selected screening protocol.

Check:

- The **number of molecules evaluated** and the **number of results** returned.
- The **score or ranking** associated with each candidate.
- How candidates are ordered and how the output can be used to select compounds for further analysis.
- **Whether similar molecules appear among the highest-ranked candidates**.
- How the **results differ between methods**, if more than one method is available.

If you compare different methods, interpret their rankings separately unless a validated strategy for combining their scores is available. Scores from different models are not necessarily directly comparable.

6) Prioritize compounds for further analysis

Use the screening results to select a manageable set of candidate molecules for subsequent investigation.

Depending on the objective, the next step may involve examining chemical diversity, predicting or inspecting binding poses, performing molecular docking, or evaluating the candidates using additional computational and experimental evidence.

HTVS is a prioritization step rather than definitive proof that a compound binds to the target.

Atomic modeling

Atomic modelling provides structural representations of biological macromolecules and their complexes. These structures can be used to investigate molecular interactions, generate hypotheses about binding, and support downstream virtual drug screening workflows.

In this practical, we will explore the atomic modelling methods available through Scipion-chem and inspect how their outputs can be used in subsequent structural analyses.

For this example there are two workflows: one where the input structure is a single chain protein, and a second one where the input structures are a protein chain and a DNA molecule.

Protein

1) Inspect the project

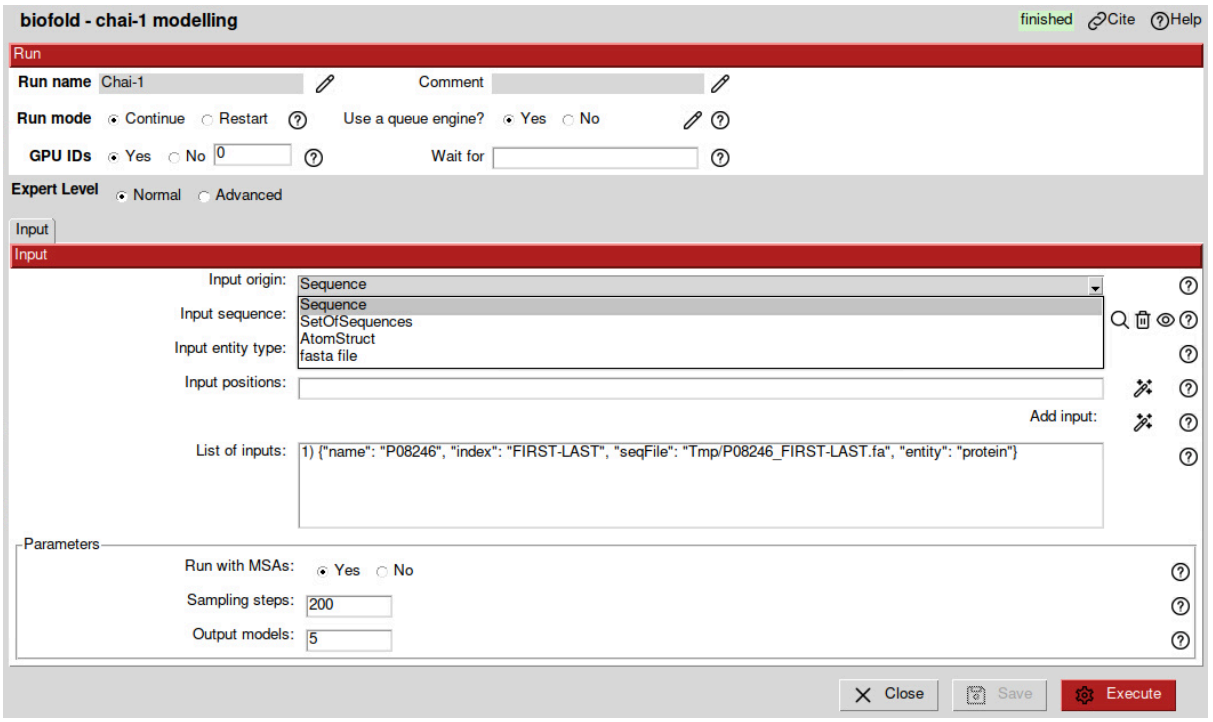
Open the project containing the workflow called "course_workflow_modeling_protein".

Identify the input data, the selected modelling protocol, and the resulting structures. In this example, the input data is a sequence object corresponding to a neutrophil elastase sequence with UniProt ID P08246.

2) Inspect the modeling protocols

There are four tools that run locally (Boltz-2, Chai-1, IntelliFold and Protenix) and an import protocol for the results obtained in the AlphaFold3 server.

The tools that run locally have as input object type “Sequence”.

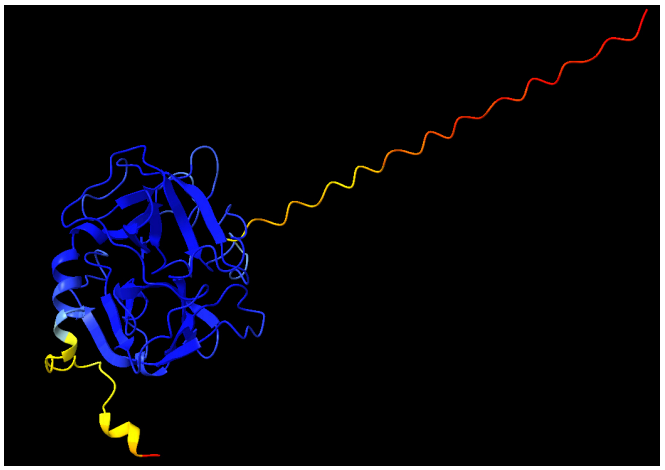


Explore the output structure using the available molecular viewers.

Check:

- Whether the **expected chains and molecular components are present.**
- Whether the **overall structure is consistent** with the intended modelling task.
- Whether the model contains the expected residues or other structural components.
- Whether confidence or quality information is available and how it is represented.
- Whether **any obvious structural problems require further investigation**.

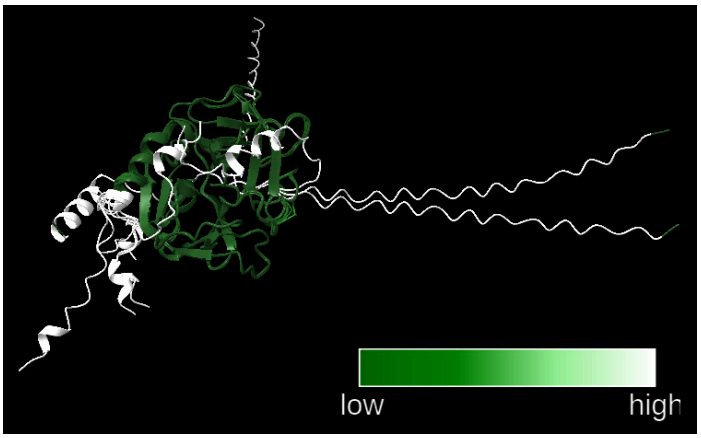
A visually plausible structure is not necessarily an accurate model. Structural interpretation should take into account the available confidence estimates, validation results, and any relevant experimental evidence.

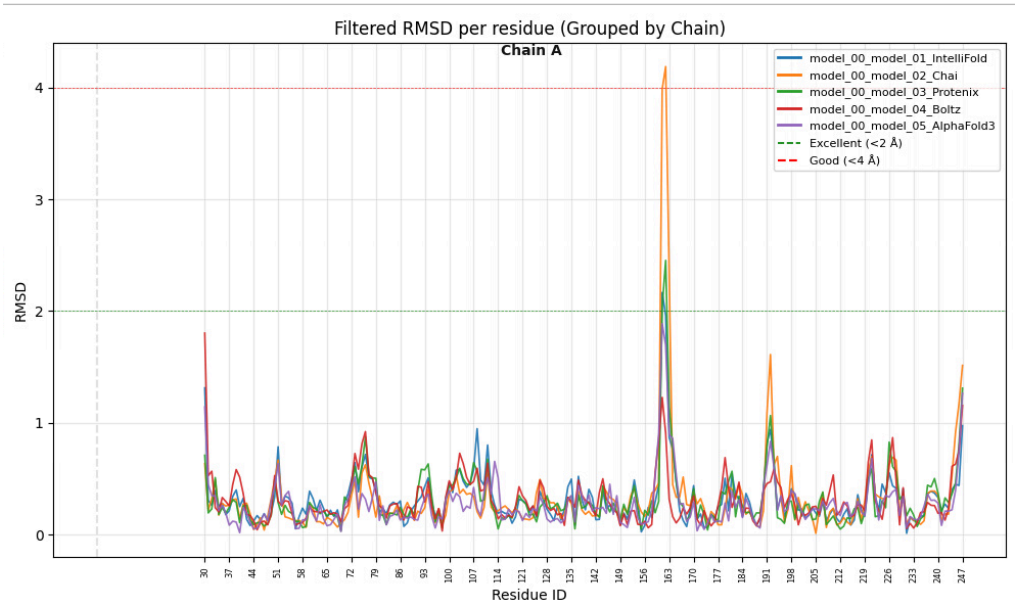


3) Inspect Chimera find discrepancies protocol

The workflow includes a protocol called “Chimera- Find discrepancies” that takes as input a reference structure as well as a list of other structures in order to compare them.

When inspecting the output, we can either view the aligned structures in chimera, each residue colored by mean RMSD with the reference, or view the RMSD graphs.





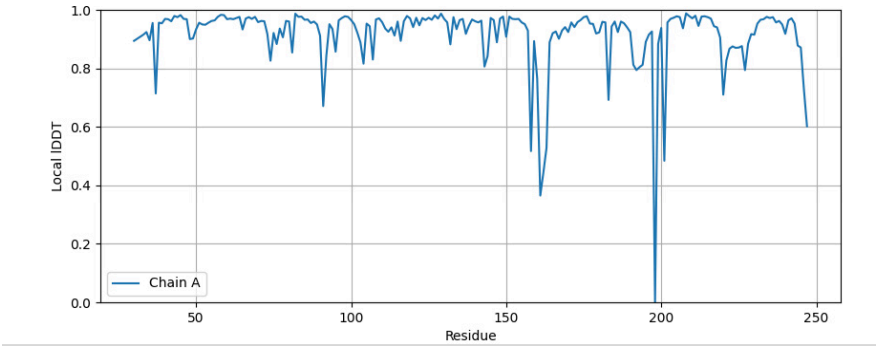
4) Inspect Openstructure results

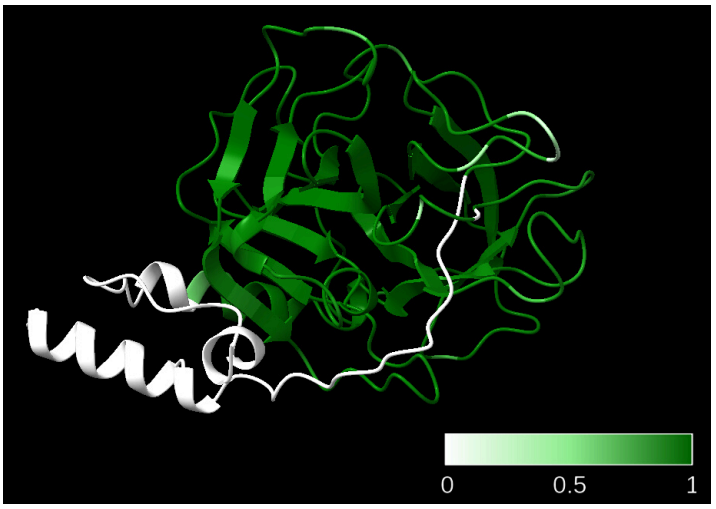
The Openstructure protocols tell us how much our predicted models differ from the reference structures obtained from the PDB.

In the summary section we can find information about different values, such as LDDT and TM-score. The viewer also lets us open the structures colored by local LDDT, as well as different informative graphs.

SUMMARY

LDDT: 0.923
Backbone LDDT: 0.983
TM-score: 0.992
QS-score (global): 1.000
DockQ (average): 0.000
DockQ (weighted): 0.000
GDT-TS: 0.992
GDT-HA: 0.968
Global RMSD (after rigid superposition): 0.545 A
Number of chains: 1





Protein-DNA complex

For the protein-dna complex the workflow is exactly the same except for the input type.

Open the project containing the workflow called "course_workflow_modeling_dna".

Identify the input data, the selected modelling protocol, and the resulting structures. In this example, the input data is an AtomStruct object corresponding to PDB 8T3X.

Protein Protein Interaction

The goal of this workflow is to analyze the interface of a protein complex, specifically an antigen–nanobody interaction. We want to improve binding by changing residues on the nanobody. In this practical, we will explore the methods available in Scipion-chem to calculate the binding energy of an interaction.

1) Inspect the project

Open the project containing the workflow called “course_workflow_protein_protein_interaction”.

In this example, the input data is a complex containing the crystal structure of the SARS-CoV-2 receptor binding domain bound to the neutralizing nanobody 4 (PDB ID: 9FC2).

2) Explore the workflow steps

2.1. Complex preparation

We are going to inspect our complex. We want two chains (antigen and nanobody) and to evaluate the interaction between them.

The target preparation protocol helps with this. With it you can remove heteroatoms and add missing residues (sometimes the structures do not contain all the residues included in the sequences). In short, we want a cleaned structure.

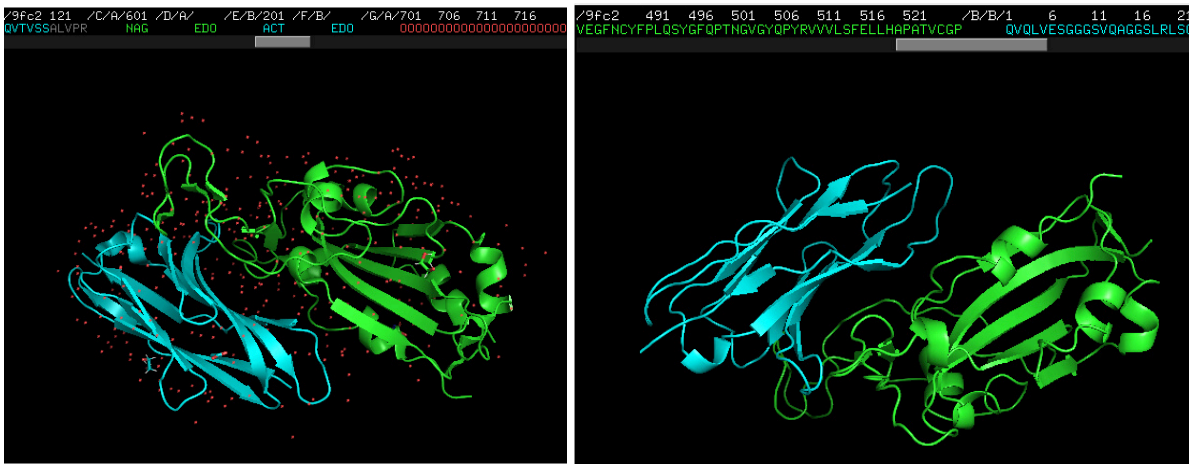
Explore the output structure using one of the available molecular viewers, such as PyMOL. Check:

- Whether the expected chains are present.
- Whether the model contains the expected residues or other structural components.
- Whether any obvious structural problems require further investigation.

We need to perform a systematic cleaning and optional repair of the input macromolecular structure to ensure compatibility with downstream computational chemistry tools.

Protocol: **pwchem – target preparation**. Available options:

- Use PDBFixer: add missing atoms, add missing residues, replace non-standard residues.
    - Remove waters.
    - Remove ligands (HETATM).
    - Select specific chains (keep chains — use the wizard).
- Comparing the output of “import atomic structure” vs. “target preparation”:



2.2. Analyze the interface

We have two protocols for this:

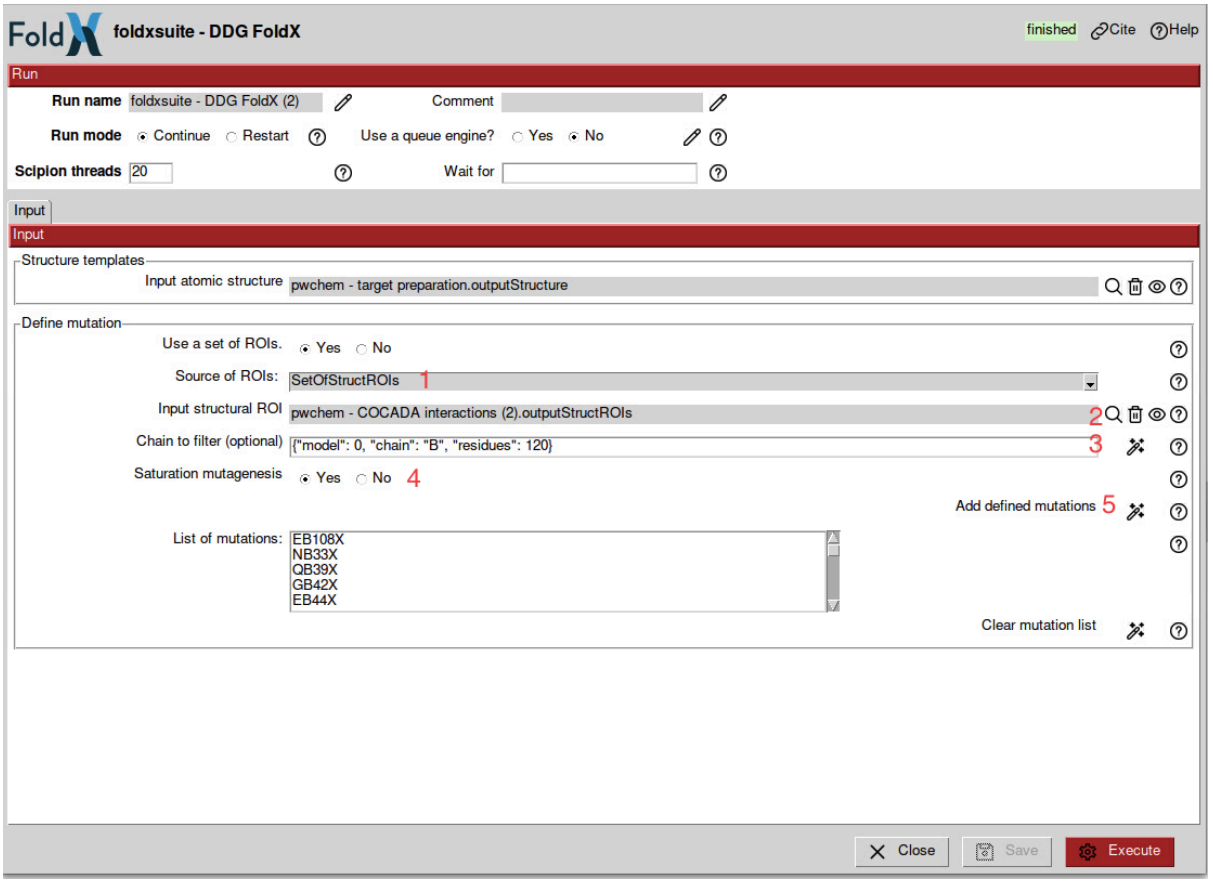
pwchem – Define structural ROIs. Here we extract ROIs from the protein–protein interface by defining an interface distance (residues within contact distance between the two protein chains). This depends on the case, but the range is normally between 3 and 5 Å.

pwchem – COCADA interactions. This produces a contact map for the input complex. The output is a set of bonds, each with its contact distance. We can keep the default values or modify them.

2.3. First energetic analysis (ΔΔG)

For this first analysis we use FoldX and SAAMBE, as they are faster tools.

As with the ROI we defined earlier, we use it to filter the residues for which we want to calculate the ΔΔG. There is also a manual option if we want to select fewer residues, and an option to choose the chain where the mutations are made. Finally, there is a saturation mutagenesis option, which mutates a position to all residues.



2.4. Ranking / Consensus

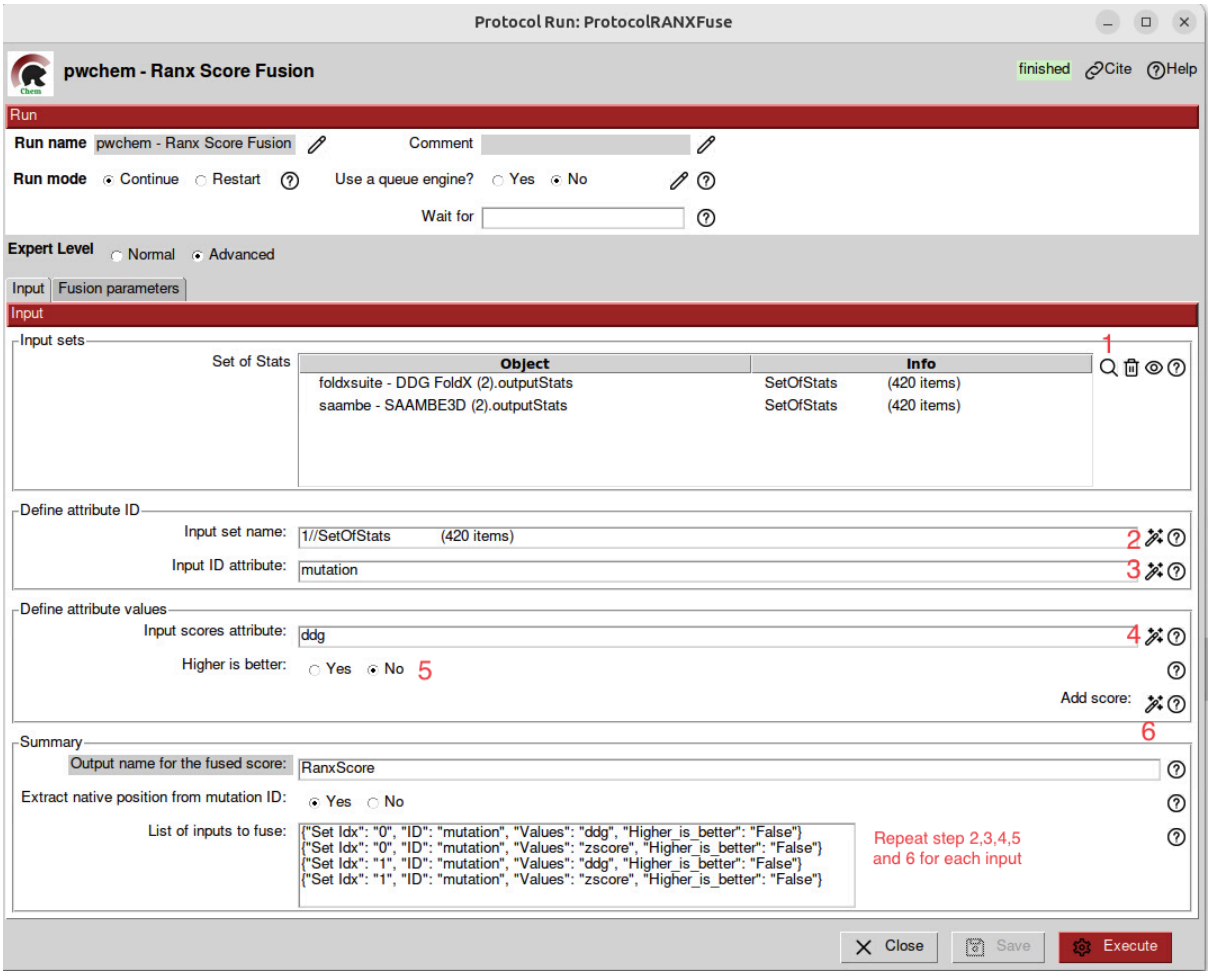
In the Rank Score Fusion protocol, select the outputs from FoldX and SAAMBE, then set:

Input ID attribute: mutation

Input scores attribute: ddg and zscore

Higher is better: No (a lower ΔΔG is better).

Extract native position from mutation ID: Yes

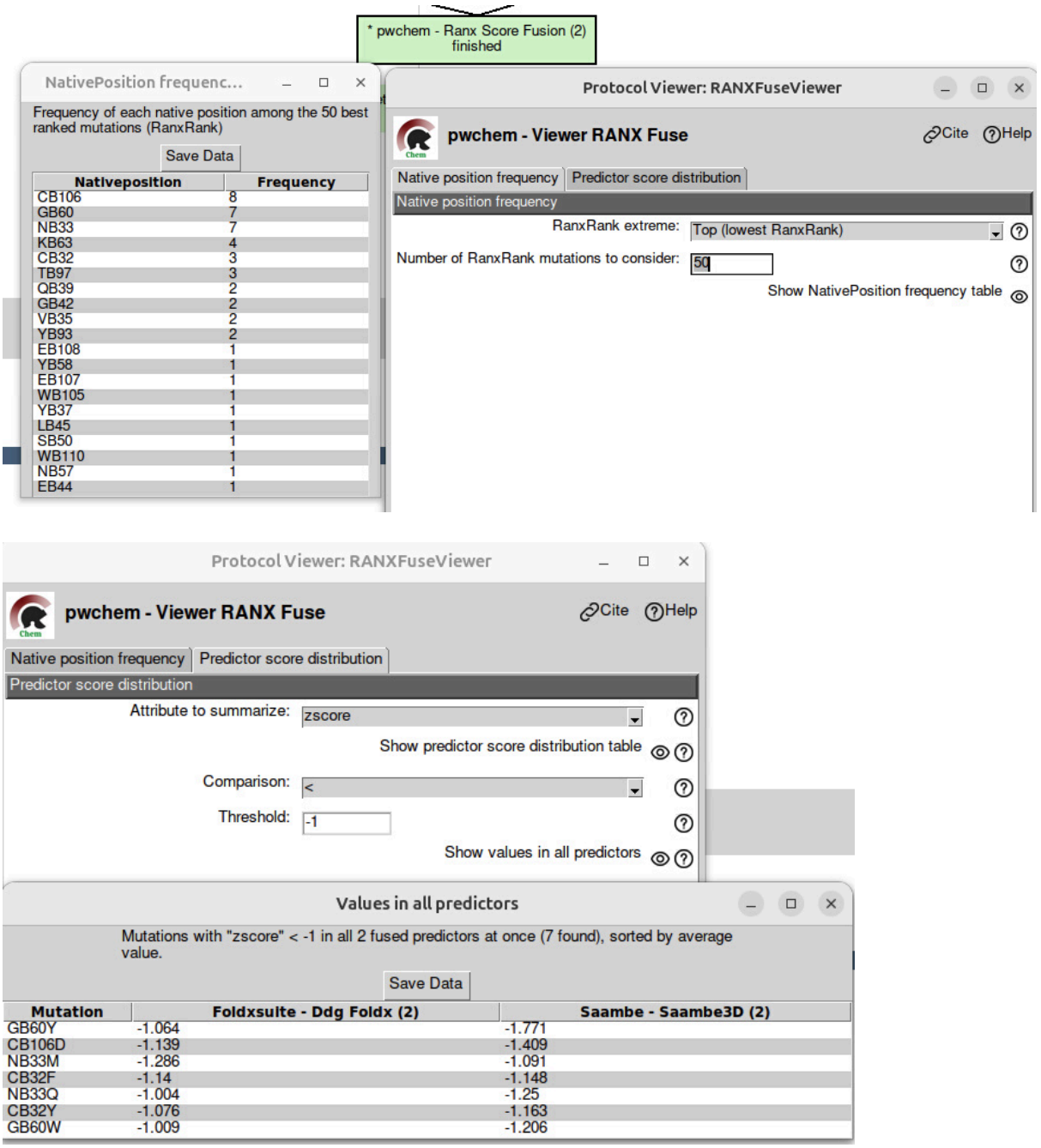


Fusion parameters:

Score-based method: med

Normalization strategy: zmuv

Analyze the output:



2.5. Second energetic analysis

Once we have the consensus of both tools (FoldX and SAAMBE), we run FlexDDG (higher computational cost) for a selected set of mutations.

Analyze the output:

	id	enabled	mutation	ddg	ddgRaw	ddgStd	nstruct	scor
1	139	☑	GB42W	-1.87	-1.93	1.48	5.0	-2.42
2	20	☑	GB60Y	-1.67	-1.47	1.53	5.0	-2.18
3	4	☑	GB60E	-1.43	-0.39	0.34	5.0	-1.89
4	95	☑	TB97R	-1.31	-0.85	1.26	5.0	-1.74
5	11	☑	GB60M	-1.19	-0.9	0.65	5.0	-1.6
6	87	☑	TB97H	-1.14	-1.16	0.32	5.0	-1.54
7	89	☑	TB97K	-1.12	-1.02	1.18	5.0	-1.52
8	2	☑	GB60C	-1.12	-0.55	0.64	5.0	-1.51
9	1	☑	GB60A	-1.11	-0.42	0.41	5.0	-1.5
10	140	☑	GB42Y	-1.06	-0.77	0.97	5.0	-1.44
11	115	☑	QB39R	-1.01	-0.39	0.36	5.0	-1.38
12	125	☑	GB42F	-1.0	-0.97	0.67	5.0	-1.37
13	127	☑	GB42H	-1.0	-1.06	0.78	5.0	-1.37
14	94	☑	TB97Q	-0.96	-0.62	0.37	5.0	-1.32
15	9	☑	GB60K	-0.92	-0.25	0.62	5.0	-1.27
16	91	☑	TB97M	-0.86	-0.92	0.74	5.0	-1.2
17	160	☑	KB63Y	-0.85	-0.99	0.78	5.0	-1.19
18	124	☑	GB42E	-0.82	-0.79	1.32	5.0	-1.15
19	5	☑	GB60F	-0.81	-0.55	0.77	5.0	-1.14
20	159	☑	KB63W	-0.8	-0.71	0.86	5.0	-1.13